

Data Analytics Case Study - Healthcare Readmissions Patient Encounters, Medication Use, and Hospital Stay

Case Study Roadmap - Ask

Guiding Questions

- What topic are you exploring?
- What is the problem you are trying to solve?
- What metrics will you use to measure your data to achieve your objective?
- Who are the stakeholders?
- Who is your audience?
- How can your insights help your client make decisions?

Key tasks

1. Identify the business task
2. Determine key stakeholders
3. Choose a dataset
4. Establish metrics

Deliverable

- ☐ A clear statement of the business task you have selected to investigate

Data Analytics Case Study

Title:	Healthcare Readmissions: Patient Encounters, Medication Use, and Hospital Stay
Industry Focus	Healthcare
Problem Statement	Hospitals need to understand patterns in patient encounters, hospital utilization, medication use, and readmissions so they can identify areas for further investigation and improve patient outcomes.
Business Use Case (What are you solving for)	1. Which five medical specialties account for the highest number of patient encounters? 2. What relationship exists between the number of medications administered and length of hospital stay? 3. How do readmission outcomes differ across payer and demographic groups? 4. Which patient and encounter characteristics are associated with readmission within 30 days?
Goals/Metric	The goal of this project is to analyze healthcare encounter data to identify patterns in patient utilization, medication use, hospital length of stay, and 30-day readmissions. The analysis will measure differences in readmission outcomes across payer and demographic groups and identify patient and encounter characteristics associated with higher recorded readmission rates.
Deliverables	Presentation: 7 slides

	Slide 1 — Title: Healthcare Readmissions & Patient Encounters Slide 2 — Business/Clinical Problem Explain why hospitals might care about: • patient volume • length of stay • medication use • Readmissions Slide 3 — Dataset & Data Preparation Show: Raw Data → Cleaning → SQL Analysis → Tableau Dashboard Slide 4 — Specialty Analysis Show your Top 5 specialties chart. Slide 5 — Medication & Length of Stay Slide 6 — Readmission Analysis Show payer + demographic comparisons. Slide 7 — Findings & Limitations Have two sections: What the data shows And What the data cannot tell us
Are datasets available	Yes
Dataset List	The dataset can be downloaded from the link given in the below section.
Websites to scrape the data if needed	https://www.kaggle.com/datasets/saurabhdayal/diabetic-patients-readmission-prediction/data?utm_source

Case Study Roadmap - Prepare

Guiding questions

- Where is your data located?
- How is the data organized?
- Are there issues with bias or credibility in this data? Does your data ROCCC?
- How are you addressing licensing, privacy, security, and accessibility?
- How did you verify the data's integrity?
- How does it help you answer your question?
- Are there any problems with the data?

Key tasks

1. Download data and store it appropriately.
2. Identify how it's organized.
3. Sort and filter the data.
4. Determine the credibility of the data. Deliverable A description of all data sources used

Deliverable

- ☐ A description of all data sources used

Data Collection & Understanding

Where is your data located?

My healthcare datasets are stored in **Google BigQuery**, where I created and organized tables for analysis. The original data was collected from publicly available healthcare data sources and imported into BigQuery for cleaning, validation, analysis, and visualization. I am using BigQuery as the primary environment for managing and querying the data.

How is the data organized?

The data is organized into separate tables based on the type of healthcare information being analyzed. Each table contains rows representing individual records and columns representing specific variables or attributes. I use SQL to examine the structure, identify relevant fields, clean the data, and connect related tables when necessary. This organization makes it possible to analyze individual variables as well as relationships between different healthcare measures.

Are there issues with bias or credibility in this data? Does your data ROCCC?

I evaluated the data using the **ROCCC** framework: Reliable, Original, Comprehensive, Current, and Cited.

- **Reliable:** I considered the reputation and purpose of the organizations that collected or published the data.
- **Original:** I reviewed the original source whenever possible rather than relying on secondary summaries.
- **Comprehensive:** I examined the available variables and records to determine whether they were sufficient to address my research question.

- **Current:** I considered the dates associated with the datasets and recognize that some healthcare data may not represent current conditions.
- **Cited:** I documented the original sources so that the data can be traced and verified.

Potential bias may exist because healthcare datasets are often collected for specific purposes, populations, or reporting requirements. Missing data, differences in reporting practices, and the populations included in the datasets could also affect the results. Therefore, I will avoid making conclusions beyond what the available data can support.

How are you addressing licensing, privacy, security, and accessibility?

I am using publicly available datasets and reviewing the source information and usage requirements before incorporating the data into my project. Because this is a portfolio project, I am avoiding personally identifiable or protected health information. I am working with aggregated or de-identified information rather than individual patient records.

I am also keeping the project data and analysis organized within Google BigQuery and limiting access to my working files and project resources. For accessibility, I will use clear chart titles, labels, readable visualizations, and explanations that allow someone without a technical background to understand the findings.

Data Integrity & Analysis

How did you verify the data's integrity?

I verified the data by examining the table structure, data types, missing values, duplicate records, and unexpected or inconsistent values. I used SQL queries to check record counts and investigate fields that could affect the analysis. I also compared results against the original source information when possible.

Before using the data for visualization, I cleaned and standardized the fields needed for analysis. I documented important cleaning and transformation steps so that the analysis can be reproduced and reviewed.

How does it help you answer your question?

The datasets provide the healthcare measures needed to investigate my research question. By combining, cleaning, and analyzing the relevant variables, I can identify patterns and relationships within the healthcare data.

Using **SQL in BigQuery** allows me to filter, aggregate, join, and transform the data. I can then use **Tableau** to create visualizations that make the findings easier to understand. Together, these tools allow me to move from raw data to evidence-based findings.

Are there any problems with the data?

Yes. The main limitations include missing or incomplete records, differences in how information may have been collected or reported, and limitations in the time period covered by the datasets.

Some variables may also require additional cleaning or transformation before they can be compared accurately.

Another limitation is that correlation in the data does not necessarily mean that one factor causes another. Because of these limitations, I will clearly document my assumptions and avoid making claims that go beyond what the data supports.

Description of Data Source

For this case study project, I used one primary dataset: **Diabetic Patients' Re-admission Prediction**, a Kaggle site registered to Saurabh Tayal.

The dataset contains healthcare information from clinical care provided at **130 U.S. hospitals and integrated delivery networks between 1999 and 2008**. It contains approximately **101,766 patient encounter records and 101 variables** related to patient demographics, hospital admissions, diagnoses, procedures, medications, laboratory testing, and readmission outcomes.

The dataset includes variables such as age, race, gender, admission type, length of hospital stay, number of laboratory procedures, number of medications, diagnoses, diabetes medications, and readmission status. The **Readmitted** variable identifies whether a patient was readmitted in less than 30 days, more than 30 days, or not recorded as readmitted.

I downloaded the dataset as a CSV file and imported it into **Google BigQuery** for data preparation and analysis. The upload proved to be difficult due to the fields containing information on medicine. The values in said fields were either NO or STEADY. Because of the multitude of NO values BigQuery was reading the field as a Boolean field instead of a String field and registering errors when encountering STEADY instead of NO. To complete the upload I used JSON text to enter the fields and values. Once the data was finally uploaded into BigQuery, I used SQL to examine the data structure, identify and address data-quality issues, transform variables when necessary, and calculate measures needed to answer my research question. I then used the prepared data to create visualizations and communicate my findings.

Although I accessed the dataset through Kaggle, the dataset documentation identifies the underlying data as the **Diabetes 130-US hospitals for years 1999–2008** dataset from the UCI Machine Learning Repository. The documentation also references the research article *Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records* by Strack et al.

Because the dataset is nearly 20 years old, and covers a specific period, hospital population, and set of variables, the findings from this project should be interpreted within the context of the available data rather than generalized to all people with diabetes or current healthcare conditions.

Case Study Roadmap - Process

Guiding questions

- What tools are you choosing and why?
- Have you ensured your data's integrity?
- What steps have you taken to ensure that your data is clean?
- How can you verify that your data is clean and ready to analyze?
- Have you documented your cleaning process so you can review and share those results?

Key tasks

1. Check the data for errors.
2. Choose your tools.
3. Transform the data so you can work with it effectively.
4. Document the cleaning process. Deliverable

Deliverable

- ☐ Documentation of any cleaning or manipulation of data

What tools are you choosing and why?

I am using **Google BigQuery, SQL, and Tableau** for this capstone project.

I chose **Google BigQuery** because it provides a structured environment for storing and managing the dataset and allows me to work with a large number of records efficiently. I am using **SQL** in BigQuery to explore, clean, transform, filter, and analyze the data. SQL also allows me to create reproducible queries that can be reviewed and modified as needed.

I chose **Tableau** for data visualization because it allows me to turn the results of my SQL analysis into interactive charts and dashboards. This will help me communicate patterns and findings in a way that is easier for a non-technical audience to understand.

Using these tools allows me to demonstrate the complete analytics process, from preparing the raw data through analysis and visualization.

Have you ensured your data's integrity?

I took steps to verify the integrity of the dataset before beginning my analysis. I examined the table structure, reviewed the column names and data types, checked the number of records, and reviewed the values contained in important fields.

I also compared the imported data with the original dataset to make sure that the data was transferred correctly into BigQuery. I checked for missing values, unexpected values, and potential duplicate records that could affect my analysis.

These checks helped me confirm that the dataset was imported correctly and that I understood the structure and limitations of the data before using it for analysis.

What steps have you taken to ensure that your data is clean?

I used SQL in BigQuery to evaluate and clean the dataset before analyzing it. My cleaning process included:

- Reviewing column names and data types.
- Checking for missing or null values.
- Identifying values that represented missing information, such as ?, where applicable.
- Checking for duplicate records.
- Reviewing categorical variables for inconsistent or unexpected values.
- Examining numerical fields for unusual or potentially invalid values.
- Standardizing fields when necessary so they could be analyzed consistently.
- Creating cleaned or transformed fields when the original format was not appropriate for analysis.
- Checking the `Readmitted` field to ensure that the outcome categories were represented consistently.

I documented the transformations and cleaning decisions so that I could explain how the raw dataset was prepared for analysis.

How can you verify that your data is clean and ready to analyze?

I can verify that the data is ready for analysis by performing quality-control checks after the cleaning process. I will compare record counts before and after cleaning, check for remaining null or unexpected values, review the data types, and confirm that important categorical fields contain the expected values.

I will also run summary queries to examine minimum and maximum values, counts, averages, and distributions for relevant numerical fields. For categorical fields, I will review the number of records in each category to identify unusual results.

Finally, I will compare selected results from the cleaned dataset with the original data to make sure that the cleaning process did not unintentionally remove or change important information.

If these checks produce expected results and the variables required for my research question are complete and consistently formatted, I can consider the data ready for analysis.

Have you documented your cleaning process so you can review and share those results?

Yes. I am documenting my data-cleaning process by saving the SQL queries I use to inspect, clean, and transform the dataset. I am organizing the queries so that I can distinguish between the original data, data-quality checks, cleaning steps, and analysis queries.

I will also document important decisions, including how I handled missing values, duplicate records, inconsistent values, and transformed variables. This documentation will make it possible for me to review my work later and explain the process to someone viewing my portfolio.

Keeping a record of my cleaning steps also makes the analysis more reproducible because another person can follow the documented process and understand how I moved from the original dataset to the data used for my analysis.

Case Study Roadmap - Analyze
Guiding questions • How should you organize your data to perform analysis on it? • Has your data been properly formatted? • What surprises did you discover in the data? • What trends or relationships did you find in the data? • How will these insights help answer your business questions?
Key tasks 1. Aggregate your data so it's useful and accessible. 2. Organize and format your data. 3. Perform calculations. 4. Document your calculations to keep track of your analysis steps. 5. Identify trends and relationships.
Deliverable ☐ A summary of your analysis

Data Organization for Analysis

I organized the data into separate stages in Google BigQuery so that the original dataset would remain unchanged while cleaned and analysis-ready versions could be created.

The data is organized into three layers:

Raw data: Contains the original Kaggle dataset as imported into BigQuery.

Cleaned data: Contains the data after reviewing data types, missing values, inconsistent values, and other data-quality issues.

Analysis data: Contains tables or queries that summarize and transform the cleaned data for specific research questions and Tableau visualizations.

This organization allows me to maintain the original source data, document my cleaning process, and create analysis tables without altering the original dataset.

Data Formatting

Yes. I reviewed and adjusted the dataset so that the fields needed for analysis were appropriately formatted in BigQuery. This included reviewing column names, data types, categorical fields, numerical fields, and missing-value indicators.

One issue I encountered during the initial upload was that BigQuery's automatic schema detection interpreted certain fields as Boolean, which resulted in import errors. I reviewed the schema and adjusted the affected fields so that the data could be imported correctly.

After the data was successfully imported, I performed additional checks to verify that the fields used in my analysis were consistently formatted and could be used in SQL queries and Tableau visualizations.

Data Discoveries

One of the first surprises I discovered was how many variables were included in the dataset. The dataset contains a wide range of information about patient encounters, including demographic characteristics, admission information, diagnoses, laboratory procedures, medications, and readmission outcomes.

I also discovered that some fields use special values such as ? to represent missing or unknown information. This required additional data-quality checks before the fields could be used confidently in analysis.

Another important discovery was that the dataset represents patient encounters rather than simply one record for every individual patient. This is important because the same patient may have multiple encounters, and I need to consider that structure when interpreting counts and readmission patterns.

These discoveries helped me better understand the limitations and structure of the dataset before drawing conclusions from the analysis.

Trends and Relationships

My analysis identified several patterns and relationships within the diabetes readmission dataset. I examined the relationship between readmission outcomes and variables such as age, admission type, length of hospital stay, number of medications, and other patient and encounter characteristics.

The analysis identified a moderate positive relationship between medication use and hospital length of stay ($r = 0.466$). Average medication use increased from 11.09 medications for one-day stays to 25.57 medications for 14-day stays. This pattern suggests that encounters involving longer hospital stays tended to involve more medications. However, the analysis identifies an association rather than a causal relationship. I also observed, contrary to my initial beliefs; that Medicaid and Medicare encounters had nearly identical recorded 30-day readmission rates. Medicaid had an 11.78% readmission rate compared with 11.75% for Medicare, a difference of only 0.03 percentage points. This suggests that the payer category alone did not substantially differentiate recorded 30-day readmission outcomes in this dataset.

Recorded 30-day readmission rates varied across race categories, ranging from 9.63% among encounters categorized as Other to 11.29% among Caucasian encounters. African American encounters had a similar rate of 11.22%, while Hispanic and Asian encounters had lower recorded rates of 10.41% and 10.14%, respectively. Because the number of encounters varies substantially across groups, particularly for smaller populations, these results should be interpreted as descriptive patterns rather than evidence of causation or medical errors.

Recorded 30-day readmission rates were similar across gender categories. Female encounters had an 11.25% recorded readmission rate compared with 11.06% for male encounters, a difference of 0.19 percentage points. The Unknown/Invalid category contained only three encounters and was excluded from meaningful comparison.

The visualizations in Tableau make these relationships easier to compare and identify. However, these relationships should be interpreted as patterns within the dataset and should not automatically be considered evidence that one variable causes another.

Recorded 30-day readmission rates varied across age groups. The 20–29 age group had the highest recorded rate at 14.24%, while the 50–59 age group had a rate of 9.67%. Among older adults, the rate increased from 11.13% for ages 60–69 to 12.08% for ages 80–89. The youngest age groups had substantially lower recorded rates, but their smaller encounter counts warrant cautious interpretation. These results identify demographic patterns for further investigation rather than establishing causal relationships.

The analysis found relatively small differences in recorded 30-day readmission rates between Medicare and Medicaid encounters and between male and female encounters. Greater variation was observed across race and age categories, particularly among age groups. These descriptive differences can help identify populations for further investigation, but the dataset does not contain enough information to determine the causes of readmission differences or evaluate specific CMS readmission-reduction programs.

As much as I began this case analysis with the belief that I might find some great disparity in the numbers surrounding the demographics of patients, specifically between patients with the ways and means to afford private pay insurance and Medicare and those on Medicaid, I am happy to understand that my assumptions were proven wrong. Encounters followed by a recorded readmission within 30 days showed higher average utilization across several measures than encounters without readmission. The 30-day readmission group averaged 4.77 hospital days, 16.90 medications, 1.22 prior inpatient visits, and 0.36 prior emergency visits, compared with 4.25 days, 15.67 medications, 0.38 prior inpatient visits, and 0.11 prior emergency visits for encounters with no readmission. These patterns indicate associations between prior healthcare utilization and recorded readmission, but they do not establish that these characteristics caused readmission.

The results are also limited to the hospitals, patient encounters, and 1999–2008 time period represented in the dataset.

Connecting Insights to Business Questions

The insights from the analysis will help answer my business questions by identifying patterns associated with hospital readmission among patients with diabetes.

By examining factors such as patient demographics, admission characteristics, length of hospital stay, medications, and other available variables, I can determine which characteristics are associated with different readmission outcomes in the dataset.

The SQL analysis provides the quantitative evidence, while the Tableau visualizations make the results easier to interpret and communicate. Together, they allow me to move from the original healthcare data to clear, evidence-based findings.

These insights can help demonstrate where patterns exist in the data and identify areas that may warrant additional investigation. Because this is an observational dataset, I will distinguish between relationships found in the data and conclusions about cause and effect.

Case Study Roadmap - Share

Guiding questions

- Were you able to answer the business question?
- What story does your data tell?
- How do your findings relate to your original question?
- Who is your audience? What is the best way to communicate with them?
- Can data visualization help you share your findings?
- Is your presentation accessible to your audience?

Key tasks

1. Determine the best way to share your findings.
2. Create effective data visualizations.
3. Present your findings.
4. Ensure your work is accessible.

Deliverable

- ☐ Supporting visualizations and key findings

Were You Able to Answer the Business Question?

Yes. The analysis allowed me to use the available patient and hospital encounter data to identify patterns associated with readmission among patients with diabetes.

Using SQL in Google BigQuery, I cleaned, organized, and analyzed the data to examine factors such as prior inpatient visits, length of hospital stay, age, admission characteristics, and other available patient and encounter variables. I then used Tableau to visualize the results.

The analysis provided evidence that helps answer the original question within the limitations of the dataset. Because the data represents historical patient encounters from a specific group of hospitals and time period, the findings should be interpreted as patterns within this dataset rather than conclusions about all patients with diabetes.

What Story Does Your Data Tell?

The data tells a story about the different characteristics and circumstances associated with hospital readmission among patients with diabetes.

The analysis shows that readmission is not represented by a single factor. Instead, the dataset contains multiple patient and encounter characteristics that can be examined together to identify patterns. Factors such as previous inpatient visits, length of hospital stay, admission characteristics, and patient demographics provide different perspectives on readmission outcomes.

The visualizations make these patterns easier to see and compare. Rather than suggesting that any individual factor causes readmission, the project uses the data to identify relationships that could be explored further.

How Do Your Findings Relate to Your Original Question?

My findings relate directly to my original question because the analysis focuses on identifying patterns in hospital readmission among patients with diabetes.

The SQL analysis allowed me to measure and compare readmission outcomes across the variables included in the dataset. The Tableau visualizations then helped communicate those comparisons in a way that makes patterns easier to recognize.

The findings provide a data-driven response to the original question while also showing the limitations of the available information. The results describe relationships found within the dataset and do not establish that one variable directly causes another.

Who Is My Audience?

My primary audience is **healthcare and business stakeholders who may need to understand patterns in hospital readmission without having extensive technical or SQL knowledge**. A secondary audience is potential employers and recruiters reviewing my data analytics portfolio.

The best way to communicate with this audience is to focus on clear findings rather than presenting large amounts of raw data or SQL code. I will use concise explanations, clearly labeled charts, key metrics, and brief descriptions of the analytical methods.

For recruiters, I will also provide access to the SQL documentation so they can review the technical process behind the analysis. This allows the same project to communicate effectively with both technical and non-technical audiences.

Can Data Visualization Help Share My Findings?

Yes. Data visualization is an important part of communicating my findings because it allows patterns and comparisons to be understood more quickly than reviewing rows of data or SQL results.

I used Tableau to create charts and dashboards that present readmission patterns visually. Bar charts allow categories to be compared, while other visualizations help show relationships and differences among patient and hospital encounter characteristics.

The dashboard brings multiple findings together in one place, allowing the audience to explore the data while maintaining focus on the original research question.

Is My Presentation Accessible to My Audience?

I am designing the presentation so that it can be understood by both technical and non-technical audiences. I will use clear chart titles, descriptive labels, readable text, and straightforward explanations of the findings.

I will avoid unnecessary technical terminology and explain important analytical concepts in plain language. The Tableau dashboard will use a consistent layout and clearly identified measures so that viewers can understand what each visualization represents.

I will also provide context for the data, including the source, time period, and limitations, so that the audience can interpret the findings appropriately. SQL queries and technical documentation will be provided separately for viewers who want to examine the analytical process in more detail.

Case Study Roadmap - Act

Guiding questions

- What is your final conclusion based on your analysis?
- How could your team and business apply your insights?
- What next steps would you or your stakeholders take based on your findings?
- Is there additional data you could use to expand on your findings?

Key tasks

1. Create your portfolio.
2. Add your case study.
3. Practice presenting your case study to a friend or family member

Deliverable

- ☐ Your top high-level insights based on your analysis
- ☐ Based on what you discover, a list of additional deliverables you think would be helpful to include for further exploration

Final Conclusion

Based on my analysis, the diabetes readmission dataset provides useful evidence of patterns and relationships among patient and hospital encounter characteristics and readmission outcomes.

My analysis using SQL in Google BigQuery and Tableau identified differences in readmission outcomes across factors such as prior inpatient visits, length of hospital stay, age, and other patient and encounter characteristics included in the dataset. These patterns provide a better understanding of which characteristics are associated with different readmission outcomes within the dataset.

However, the analysis does not establish that any individual factor causes hospital readmission. The dataset represents patient encounters from 130 U.S. hospitals and integrated delivery networks during 1999–2008, so the findings should be interpreted within that population and time period.

Overall, the project demonstrates how data cleaning, SQL analysis, and data visualization can be used together to identify patterns in healthcare data and communicate those findings to stakeholders.

How Could Your Team and Business Apply Your Insights?

Healthcare teams could use the patterns identified in this analysis as a starting point for asking more detailed questions about hospital readmission.

For example, the findings could help stakeholders identify patient or encounter characteristics that are associated with higher or lower readmission rates within the dataset. These patterns could then be used to determine where additional investigation may be useful.

The analysis could also help demonstrate how healthcare organizations can use data to monitor outcomes, identify patterns, and support evidence-based decision-making. However, the findings should be considered exploratory and should be validated with more current and representative data before being used to make operational or clinical decisions.

What Next Steps Would You or Your Stakeholders Take?

The next step would be to validate the patterns identified in this project using more recent and comprehensive healthcare data.

Stakeholders could investigate the factors that showed notable relationships with readmission in greater detail and determine whether those patterns remain consistent across different hospitals, patient populations, and time periods.

Additional analysis could also examine whether multiple factors occurring together are associated with different readmission outcomes. Statistical analysis or predictive modeling could be considered in a future project if appropriate data were available.

Before applying any findings operationally, stakeholders should validate the results and consider clinical context, data quality, privacy requirements, and other factors that are not represented in the current dataset.

Additional Data That Could Expand the Analysis

Yes. Additional and more recent data could provide a broader understanding of diabetes-related hospital readmissions.

Potential data sources or variables could include:

- More recent hospital encounter data to determine whether the patterns have changed over time.
- Additional hospitals and healthcare systems to increase the representativeness of the analysis.
- Patient socioeconomic information, such as insurance status or socioeconomic indicators.
- More detailed information about patients' medical histories and chronic conditions.
- Follow-up care after discharge, including outpatient visits and care-management programs.
- Information about medication adherence and changes in treatment.
- Additional clinical measures, such as laboratory results and other indicators of patient health.
- Social determinants of health that may influence access to healthcare and follow-up services.

Adding these types of information could help provide more context around the patterns identified in the current analysis. It could also support more advanced analysis while providing a stronger basis for understanding factors associated with readmission.